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Contamination fear and attention bias variability early in the COVID-19 pandemic

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ABSTRACT

The onset of the COVID-19 pandemic resulted in a dramatic increase in the salience and importance of information relating to both the risk of infection, and factors that could mitigate against such risk. This is likely to have contributed to elevated contamination fear concerns in the general population. Biased attention for contamination-related information has been proposed as a potential mechanism underlying contamination fear, though evidence regarding the presence of such biased attention has been inconsistent. A possible reason for this is that contamination fear may be characterised by variability in attention bias that has not yet been examined. The current study examined the potential association between attention bias variability for both contamination-related and mitigation-related stimuli, and contamination fear during the early stages of the COVID-19 pandemic. A final sample of 315 participants completed measures of attention bias and contamination fear. The measure of average attention bias for contamination-related stimuli and mitigation-related stimuli was not associated with contamination fear ($r = 0.055$ and $r = 0.051$, $p > 0.10$), though both attention bias variability measures did show a small but statistically significant relationship with contamination fear ($r = 0.133$, $p < 0.05$; $r = 0.147$, $p < 0.01$). These attention bias variability measures also accounted for significant additional variance in contamination fear above the average attention bias measure (and controlling for response time variability). These findings provide initial evidence for the association between attention bias variability and contamination fear, underscoring a potential target for cognitive bias interventions for clinical contamination fear.

Contamination fear describes the excessive fear of becoming contaminated by germs, dirt, or other contaminants and is a major fear subtype of obsessive-compulsive disorder (OCD), being present in around 50% of OCD cases (Rasmussen & Eisen, 1992). Heightened contamination fear is characterised by persistent and excessive fear of harmful contaminants, or a fear of causing harm to oneself or others through contact with contaminants (American Psychiatric Association, 2013). These fears can significantly disrupt daily activities and cause distress. People with contamination fears may also have other obsessive-compulsive symptoms, such as excessive checking, counting rituals, and in particular washing rituals that can contribute to considerable functional impairment (Rachman, 2004).

The onset of the COVID-19 pandemic resulted in a dramatic increase in the salience and importance of personal hygiene and cleanliness, as well as legitimate concerns about the spread of the virus. For individuals with already elevated contamination fear, including those with clinical OCD diagnoses, the COVID-19 pandemic has been particularly challenging (Fontenelle et al., 2021). The focus on hand-washing, avoiding surfaces, not touching the face, wearing masks, coupled with media coverage of the virus, also serve as reminders and cues for people with contamination fears, leading to increased anxiety (Moghanibashi-Mansourieh, 2020). Particularly at the early stages of the pandemic, the saturation coverage likely contributed to heightened awareness and concern about germs and infection. Indeed, evidence suggests that since

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the onset of the pandemic there has been a significant increase in obsessive-compulsive washing symptoms as compared to pre-pandemic levels (Knowles & Olatunji, 2021).

While extreme levels of contamination fear can contribute to distress, as demonstrated by the COVID-19 pandemic, it is also the case that concerns regarding contamination can be highly adaptive. Disgust and fear in response to sources of potential contamination confer an evolutionary advantage, allowing us to avoid potential sources of ill health, suggesting that non-clinical levels of contamination fear may play a functional role in the avoidance of disease (Curtis et al., 2011; Verwoerd et al., 2013). As such the ability to identify such threats and mitigate against them, while not exacerbating anxiety and behavioural symptoms to clinical levels, can be essential to maintaining emotional and physical wellbeing.

Understanding cognitive processes that underpin the identification of, and response to contamination-related threats can potentially inform the adaptive expression of contamination fear and associated behavioural responding. Biased attention to threat has been consistently implicated across many clinical and sub-clinical manifestations of anxiety (Bar-Haim, 2010) including contamination fear. A number of studies have found evidence consistent with the presence of an attention bias favouring contamination-related information. For instance, in an early study in a very small sample (n = 13) of individuals diagnosed with OCD who reported significant contamination concerns, Tata et al. (1996) found that those with OCD showed greater attentional vigilance for contamination-related concerns as compared to a comparison group with general high trait anxiety (n = 18). Similarly, in a study comparing an elevated contamination fear group (n = 23) to a low contamination fear control group (n = 28), Cisler and Olatunji (2010) observed that the contamination fear group showed delayed attention disengagement from disgust-related stimuli compared to neutral stimuli, which was not evident in the control group, suggesting greater attentional capture by disgust-related information.

Research examining biased attention in contamination fear and OCD has been far from consistent, however. In a study employing an eye-tracking task, Armstrong et al. (2012) found evidence that participants high in contamination fear (n = 19) oriented gaze toward contamination threat stimuli more often than participants low in contamination fear (n = 20). However, in contrast to Cisler and Olatunji (2010), the contamination fear group fixated for *shorter* durations on contamination fear stimuli as compared to other information, a pattern the authors speculated could relate to vigilance followed by subsequent avoidance. Bradley et al. (2016) found no evidence for an association between faster attentional orienting to obsessive-compulsive related stimuli (including washing) and obsessive-compulsive symptom severity, but did find an association with longer fixation durations. In a study examining high (n = 23) and low contamination fear (n = 25), Armstrong et al. (2010) found no evidence for greater attentional orienting to disgust-related stimuli but did find evidence for maintenance of attention to both disgust stimuli and general threat stimuli compared to the low contamination fear group. These findings suggest that while biased attention in some form may be a feature of OCD/contamination fear, there is relatively little consistency with regard to the precise manifestation of this bias.

Vogt et al. (2011) first identified the potential relevance of selective attention to information associated with the mitigation of disgust-related threat by incorporating stimuli representing cleanliness. In their study, unselected participants were allocated to either a disgust-induction procedure (n = 20; touching fake disgust-related objects) or a control procedure (n = 19; touching non-disgust inducing objects) and subsequently completed an attention bias assessment task. They found that while participants in both conditions attended more to disgust-related stimuli overall, those in the disgust-induction condition showed greater attentional vigilance for mitigation-related information (cleanliness stimuli) as compared to the control condition. Therefore, while the study did not explicitly examine attention bias as a function of contamination fear, the findings suggest that selective attention may be dynamically shaped by proximity/exposure to contamination threat. The study also critically highlights the potential importance of including mitigation-related information in research examining patterns of biased contamination fear, which can inform adaptive patterns of biased attention in relation to contamination threat. This may be particularly relevant when studying attention bias in the context of situations involving real/perceived threat of contamination.

As suggested by the findings of Armstrong et al. (2012), a further reason for potential inconsistency in findings is that heightened contamination fear may be characterised by variation between attentional vigilance towards contamination threat, and also attentional avoidance of such information at different times. Such variation between vigilance and avoidance over time would essentially be exhibited as higher variability in attention bias both toward and away from contamination threat. Indeed, the possibility that some forms of emotional pathology may be underpinned by attention bias variability (ABV) has been a growing area of research interest. Iacoviello et al. (2014) was among the first to examine the presence of ABV in post-traumatic stress disorder (PTSD). Noting inconsistent findings in studies examining attention bias in PTSD, the researchers speculated that heightened PTSD symptomatology may be characterised by dysregulation of attention to threatening information. They implemented an assessment of ABV derived from the standard dot probe task and examined variance in attention bias over time, finding that those with PTSD showed higher ABV compared to controls and trauma-exposed individuals without PTSD. Recent meta-analytic findings have suggested preliminary evidence for ABV in anxiety and depression, and consistent evidence in relation to PTSD (Todd et al., 2022), with some findings consistent with a potential causal role in PTSD (Todd et al., 2023). To date however only a single study has sought to examine whether ABV is a feature of heightened contamination fear. Specifically, Todd et al. (2021) found that in a between-group design comparing high (n = 31) and low (n = 44) contamination fear groups found that, regardless of group, distress experienced from intrusive thoughts following a lab-based contamination stressor was predicted by attention bias variability, with some evidence that standard average attention bias also predicted intrusion distress (for the high contamination fear group only). This research therefore provides some preliminary support for the role of ABV in contamination fear, albeit in a lab-based context and without consideration of the potential role of attention for mitigation-related information.

A further reason for increasing interest in the construct of attention bias variability has been the recognition of the poor reliability of the standard average measures of attention bias. More frequent reporting of reliability measures for the standard average measure of attention bias derived from the probe task led to the recognition that the task generally has poor internal consistency (cf. Parsons et al., 2019). A number of subsequent studies have since noted higher levels of reliability on attention bias variability measures compared to the standard probe task index measures. Among the first to examine ABV, Zvielli, Bernstein, and Koster (2015) reported very low reliability for the traditional average bias measure (r = 0.06) and comparatively superior (though still limited) reliability of the ABV measure (r = 0.46). Clarke et al. (2020) similarly found higher reliability in the ABV measure (r = 0.63) compared to the average bias measure (r = 0.18), a pattern highly consistent with Carlson and Fang (2020) who observed a higher average internal consistency for ABV measures (r = 0.58) compared to average attention bias measures (r = 0.18). Thus, an additional attraction of ABV research is the comparative improvement in reliability beyond the very low levels observed in traditional measures, albeit with levels of internal consistency that may still not fully meet traditional standards of acceptability in individual differences research.

Low levels of reliability on traditional average attention bias measures may also provide an explanation for inconsistent findings in past research. Specifically, measures that possess low reliability tend to produce wider distributions of scores which can increase the likelihood of sampling error withing low-powered studies, increasing the likelihood of false positive results (Loken & Gelman, 2017; Schmidt et al., 2014). Given that much of the research examining patterns of biased attention in contamination fear has been conducted with small samples, effects reported from such low powered studies should be regarded with appropriate scepticism.

While ABV has been a construct of increasing interest for cognitive bias researchers, concerns have been raised about the possibility that effects may be attributable to a confounded process, namely response-time variability. An original study using modelled data by Kruijt et al. (2016) highlighted that a measure of ABV derived from the attentional probe task will be the product of both variability in attention bias, and variability in response times. This highlighted the importance of controlling for general response time variability when assessing relationships involving ABV. For the current study, this was achieved using a task with the same parameters as the standard attention probe, though with non-emotional stimuli to yield an isolated measure of response-time variability as a control variable in examining relationships involving ABV.

Given inconsistent findings and the relative neglect of mitigation-related information when assessing attention bias in contamination fear, the present study sought to address outstanding research questions in relation to these issues during a time of heightened contamination threat in the general population. Specifically, the aim was to determine whether a relationship exists between ABV, for both contamination-related and mitigation-related stimuli, and contamination fear, while controlling for response-time variability. Furthermore, to the extent that ABV is associated with contamination fear, we also undertook to determine if ABV measures account for significant unique variance above standard average attention bias to contamination/mitigation-related information. Given that past inconsistencies may also be due to relatively low power, these patterns of bias were examined in a large cross-sectional sample. The onset of the COVID-19 pandemic provided a unique opportunity to examine relationships involving vigilance for both contamination-related and mitigation-related information at a period in time when the threat value of this information was elevated, increasing the salience of such information along with and the objective benefits of identifying sources of contamination threat, and mitigation information. The present study was conducted between April and June 2020, a period corresponding with the initial relaxation of some COVID restrictions in the study's jurisdictions. This meant that many residing in these areas were beginning to resume some community-based activities, albeit with considerable restrictions and many reminders of risks in place (e.g. mask mandates, digital check-in at venues etc.) meaning potentially heightened salience of contamination threat and methods of mitigation.

1. Method

1.1. Participants

An initial sample of 347 participants were recruited from undergraduate samples and social media advertising. Participants were recruited via respective undergraduate participant pools and community volunteers via Curtin University, the University of Sydney, the University of Western Australia, and the University of Sussex. This project was conducted as part of the Cognition and Emotion Research Collaboration (CERCI) network which seeks to pool research resources to increase power and enhance the reliability of research findings in cognition and emotion. Following data cleaning and exclusion of outliers (see Results section) the final sample comprised 315 participants, 225 female, 87 male, and 3 non-binary with a mean age of 20.80 (SD = 4.99). A sensitivity power analysis for regression using G*Power with $\alpha = 0.05$ and power at 80% indicated that this sample size was sufficient to detect small effects ($f^2 = 0.03$). Full descriptives by institution are provided in Table 1. All recruitment and testing procedures were approved by the respective institution's human research ethics committee and were in line with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Helsinki Declaration of 1975 (as revised in 2000). All participants provided informed consent prior to participation.

1.2. Assessment of contamination fear (Padua Inventory)

We employed the 10 item Contamination Obsessions and Washing Compulsions subscale of the Padua Inventory (PI–COWC) to measure participants' levels of contamination fear (Burns et al., 1996). The scale assesses the degree to which an individual relates to symptoms of contamination fear, for example, "I find it difficult to touch an object when I know it has been touched by strangers or by certain people". Scores range from 0 ("Not at all") to 4 ("Very much") and are summed, resulting in a total subscale score. Consistently high levels of reliability have been observed for this subscale (Rubio-Aparicio et al., 2020). Internal consistency was high in our sample (Cronbach's alpha = 0.914).

1.3. Assessment of attention bias and attention bias variability

The standard measure of average attention bias and attention bias variability were derived from an attention probe assessment task. Stimuli and task structure are detailed below.

1.3.1. Task stimuli

To assess attention bias and ABV for both contamination-related and mitigation-related information, the task required contamination-related vs neutral word pairs and mitigation-related vs neutral word pairs. To attain this, an initial list of candidate words comprising 80 neutral words, 53 words relating to contamination and 42 words related to mitigation of contamination were generated by the research team and subsequently rated by twelve independent judges. Candidate words were rated on a 9-point scale in relation to their degree of relatedness to virus concerns (1 = "Highly related", 5 = "Moderately related, 9 = "Completely unrelated"). Words rated 5 or lower on this scale were then rated on their relatedness to either contamination risk or protection from contamination (1 = "Strongly related to risk", 5 = "Related to both", 9 = "Strongly related to protection"). All words were also rated on familiarity ("How familiar is this word to you?") on a 9-point scale (1 = "Highly unfamiliar" to 9 = "Highly familiar"). The top 24 words rated most relevant to contamination risk, and 24 words related to protection were selected and paired with a neutral word (rated as being more unrelated to virus concerns: above 7) that were equal in character length and with the most similar familiarity ratings. This resulted in 24 contamination-neutral word pairs (e.g. 'virus' - 'click') and 24 mitigation-neutral word pairs (e.g. 'sanitizer' - 'dialogues').

1.3.2. Task structure

Each trial of the probe task commenced with a fixation cross in the centre of the screen for 500ms. A word pair (either contamination-neutral or mitigation-neutral) then appeared, vertically aligned 25 mm apart, one above and one below the position of the fixation cross for 500ms. The word pair was then replaced by the target probe comprising two red dots aligned either vertically (':') or horizontally ('‥') appearing in the location vacated by the top or bottom word. Participants were instructed to identify the target probe by pressing the number 1 for vertical dots or 2 for horizontal dots, as quickly as possible without compromising accuracy. A correct response cleared the screen and the next trial commenced after a 500ms inter-trial-interval. An incorrect response triggered a 3-s delay and the message 'error' in the centre of the screen. The target probe replaced the neutral word and the salient stimulus word (contamination-related or mitigation-related) with equal frequency across trials. The word pairs were presented in randomised blocks of 48 trials. The position of the neutral-salient stimulus pairs

Table 1

Demographic data (participant number, gender, mean age and DASS-21 score) and mean values for predictor variables and criterion (Padua Inventory) variable with F values and significance for one-way ANOVAs across institutions. Standard deviations given in parentheses.

	Institution					
	Curtin University	UWA	University of Sydney	University of Sussex	Total sample	
Participants N	97	66	108	44	315	
% of total sample	30.8	20.9	34.3	14.0	100	
Gender						
Female	75	39	75	36	225	
Male	22	26	31	8	87	
Non-binary	0	1	2	0	3	
Age Years	21.1 (4.8)	21 (6.92)	20 (2.84)	22 (5.9)	20.8 (4.99)	
					F	***p***
RTV	137.89 (70)	171.67 (221.53)	154.58 (175.02)	130.12 (83.55)	0.92	0.431
ABI_{Contam}	−0.83 (63.58)	−3.66 (35.59)	7.67 (45.2)	−6.01 (47.17)	1.49	0.221
ABI_{Mit}	−0.72 (56.07)	−5.49 (56.81)	5.53 (52.97)	3.97 (43.87)	0.64	0.592
$TLBS_{Contam}$	178.81 (68.15)	170.66 (66.78)	162.33 (58.64)	166.87 (62.64)	1.16	0.329
$TLBS_{Mit}$	170.55 (65.99)	162.74 (63.04)	170.99 (72.28)	164.92 (63.21)	0.30	0.829
Padua-Contam	13.03 (8.65)	11.83 (9.97)	16.47 (9.4)	14.57 (9.76)	3.91	0.010

Note: ABI_{Contam} = average attention bias index measure for contamination-related stimuli; ABI_{Mit} = average attention bias index measure for mitigation-related stimuli; ABV-Mov_{Contam} = Moving average measure of attention bias variability for contamination-related stimuli; ABV-Mov_{Mit} = Moving average measure of attention bias variability for mitigation-related stimuli; $TLBS_{Contam}$ = Trial-level bias score measure of attention bias variability for contamination-related stimuli; $TLBS_{Mit}$ = Trial-level bias score measure of attention bias variability for mitigation-related stimuli; RTV = response time variability measure; Padua-Contam = Contamination scale of Padua Inventory. The full dataset for the current paper is available at: https://osf.io/rvczy/.

(upper/lower), and the probe location (upper/lower) were counterbalanced across blocks, and the probe type was randomised on each trial within the constraint that an equal number of each probe type appeared within each block of 48 trials. Participants completed four blocks for a total of 192 trials.

1.3.3. *Measures of attention bias*

Three measures of attention bias were computed from the above task for each salient stimulus type (contamination and mitigation). The standard measure of average attention bias to the salient stimulus over neutral stimuli was computed by subtracting average response time latencies on trials where the probes replaced the salient stimulus (salient-stimulus trials) from average latencies on trials where probes replaced the neutral stimulus (neutral-stimulus trials). The resulting attention bias index (ABI) was computed for both contamination related-stimuli (ABI_{Contam}) and mitigation-related stimuli (ABI_{Mit}).

Two different measures of attention bias variability have been commonly employed in the literature. As highlighted in a recent meta-analysis (Todd et al., 2022), few studies have included both measures, meaning that there has been limited ability to compare and contrast the relative sensitivity of each. As such, we elected to include both measures in the current study. The first of these was the moving average measure of attention bias variability (ABV-Mov) based on the methodology developed by Naim et al. (2015). This approach involves the initial computation of multiple attention bias index scores by subtracting individual salient-stimulus trial latencies from individual neutral-stimulus trial latencies for each participant, according to their temporal appearance in the task. Multiple moving averages of these bias index scores are then computed for groups of 10 index scores (i.e. index 1–10, 2–11, etc.). The final ABV-Mov measure is computed as the standard deviation of the moving average scores, divided by the participant's overall mean reaction time. This measure was computed for contamination-related stimuli (ABV-Mov_{Contam}) and mitigation-related stimuli (ABV-Mov_{Mit}).

The second measure of attention bias variability was based on the trial level bias score (TLBS) approaches employed by Clarke et al. (2020), in turn based on the approach developed by Zvielli, Amir, et al. (2015). In this approach, trials are initially paired according to their temporal order of appearance on the task. Specifically, the response latency for the first probe-neutral trial is paired with the response latency for the first presented probe-salient trial, the second presented probe-neutral trial is paired with the second presented probe-salient trial and so on. Individual response latencies for probe-salient trials are then subtracted from individual probe neutral trials resulting in a temporal sequence of attention bias index scores. The difference between each index score and the subsequent index is then computed (i.e. index1 - Index2, Index2 - Index3, etc.), the values are normalised, summed, and divided by the number of observations to achieve the final index of TLBS. This measure was computed for contamination related-stimuli ($TLBS_{Contam}$) and mitigation-related stimuli (TLBS-Mov_{Mit}).

1.4. *Assessment of response time variability*

Researchers have employed different approaches in controlling for response-time variability across past studies (Carlson et al., 2022; Clarke et al., 2020; Swick & Ashley, 2017; Takano et al., 2021). In selecting a method to control for the effect of response time variability, we were guided by the findings of Kruijt et al. (2016) and the specific principle that response times in the standard probe task will be influenced by variability in attentional bias and general variability in response times. As such, it is not possible to derive an isolated measure of response time variability from trials that also seek to assess attention bias, because responses on these trials will reflect both response time variability and attention bias variability to the different stimulus types. Therefore, to control for variance in response times in isolation, it is necessary to precisely mimic the trial structure of the task, though without emotionally valenced stimuli that will elicit bias variability.

Thus, the structure of the RTV assessment task was identical to the attentional probe task with the exception of the stimuli used. In place of salient-neutral stimulus pairs, the task instead included paired character strings. Specifically, the stimuli comprised 24 paired strings of X's. To ensure perceptual similarity between the task parameters, the length of the strings were matched to the word stimuli on the attention bias assessment probe task. For example, the word pair PATHOGEN – ANYTHING would be replaced with the length-matched string XXXXXXXX – XXXXXXXX. The RTV assessment task comprised 96 trials delivered in a separate block to the attention probe assessment task. Probe type and position on each trial were randomised within each block of 24 trials. The standard deviation of response times across all trials of the task was taken as the measure of RTV.[1]

1.5. Procedure

Participants were undergraduate students and members of the general community that were recruited either via undergraduate participant pools or social media advertising. The study was hosted on Inquisit 5 online (Millisecond Software, 2018) with participants completing the study remotely. Participants were first presented with study information and provided informed consent. An initial screen calibration routine ensured that visual parameters remained constant across different screen sizes and resolutions. Participants then completed demographic information, questionnaires including the contamination subscale of the Padua Inventory. Participants then completed the attention probe assessment task which included detailed instructions on task completion and an initial 12 practice trials with neutral-neutral stimuli, followed by the response time variability assessment task. At the conclusion participants were debriefed as to the purpose of the study. The total study took approximately 30 min to complete.

2. Results

2.1. Data preparation

Preparation of response time data from the probe and response time variability tasks involved initial exclusion of incorrect responses and individual response times falling 3 median absolute deviations (Leys et al., 2013) from each participants median reaction time for each trial type. This resulted in the exclusion of 13.86% of trials in total. While broadly consistent with rates of exclusion in similar large-scale online studies (Clarke et al., 2020 12.48%), this is slightly higher as compared to lab-based studies (Basanovic et al., 2017; Notebaert et al., 2015; Notebaert et al., 2016 7.58%, 8.7%, and 8.95% respectively). We adopted an inclusion criterion of at least 70% accuracy on the probe task. Thirteen cases (3.7%) were excluded on the basis of this criterion. Given the use of word stimuli, participants reporting low fluency in English were excluded resulting in the loss of seven participants (2.1%). Finally, any participant whose scores on any of the attention bias measures fell further than 3 SD from the mean value were also excluded. This resulted in the exclusion of 12 participants (3.7%). This resulted in a final sample of 315 participants. Table 1 summarises sample characteristics.

Split-half (first/second half of trials) correlations were conducted to examine internal consistency of each bias index measure. In line with other studies (Chapman et al., 2019; Schmukle, 2005), these reliability levels were low for the average attention bias index measures: $\mathrm{ABI_{Contam}}$ $r(315) = 0.236$, $p < 0.001$ (Spearman-Brown corrected $r = 0.382$); $\mathrm{ABI_{Mit}}$ $r(315) = 0.246$, $p < 0.001$ (Spearman-Brown corrected $r = 0.395$), and similarly low for the moving average attention bias variability measures: $\mathrm{ABV\text{-}Mov_{Contam}} = r(315) = 0.254$, $p < 0.001$ (Spearman-Brown corrected $r = 0.405$); $\mathrm{ABV\text{-}Mov_{Mit}} = r(315) = 0.177$, $p = 0.002$ (Spearman-Brown corrected $r = 0.301$). The trial-level bias score measures were observed to have higher reliability: $\mathrm{TLBS_{Contam}}$ $r(315) = 0.582$, $p < 0.001$ (Spearman-Brown corrected $r = 0.736$); $\mathrm{TLBS_{Mit}}$ $r(315) = 0.609$, $p < 0.001$ (Spearman-Brown corrected $r = 0.757$), while the measure of response time variability showed very high split-half reliability: RTV $r(315) = 0.900$, $p < 0.001$ (Spearman-Brown corrected $= 0.947$). To further examine the potential internal consistency of the ABV measures when controlling for response-time variability, we also examined partial the split-half associations controlling for RTV. The pattern of the associations were similar with slightly reduced values for moving average attention bias variability measures: $\mathrm{ABV\text{-}Mov_{Contam}} = r(315) = 0.243$, $p < 0.001$ (Spearman-Brown corrected $r = 0.391$); $\mathrm{ABV\text{-}Mov_{Mit}} = r(315) = 0.167$, $p = 0.003$ (Spearman-Brown corrected $r = 0.286$) and the trial-level bias score measures: $\mathrm{TLBS_{Contam}}$ $r(315) = 0.564$, $p < 0.001$ (Spearman-Brown corrected $r = 0.721$); $\mathrm{TLBS_{Mit}}$ $r(315) = 0.589$, $p < 0.001$ (Spearman-Brown corrected $r = 0.741$). Within the context of individual difference research this places the TLBS measures within the 'acceptable' level according to the Spearman-Brown corrected values, and all other measures well below the acceptable range.

2.2. Relationships between variables

To examine relationships between variables and to determine if either or both ABV measures were associated with contamination fear, bivariate correlations were examined. As summarised in Table 2, this showed that the only attention bias measures to demonstrate a significant relationship with contamination fear were the $\mathrm{TLBS_{Contam}}$ and $\mathrm{TLBS_{Mit}}$ measures. RTV showed significant correlations with all attention bias variability measures, highlighting the relevance of including this as a covariate. As neither of the ABV-Mov measures were related to contamination fear, these measures were not considered in subsequent analyses.

2.3. Characteristics across recruitment sites

Due to the complexities of comparison of categorical variables with more than two categories in regression models the effects of 'Institution' are summarised in initial ANOVA comparison and subsequent t-tests. Full details of all analyses and models are accessible via https://osf.io/rvczy/. Comparison of predictor variables and contamination fear by recruitment site using one-way ANOVAs (Welch's) revealed a significant effect for contamination fear only $F(3,137) = 3.91$, $p = 0.010$, while none of the other measures approached significance (see Table 1). Games-Howell post-hoc t-tests revealed that this effect was due to differences in mean Padua scores between the Curtin and University of Sydney recruitment sites $t(1,202.9) = 2.37$, $p = 0.035$, $M_{diff} = 3.44$, and between the University of Sydney and the University of Western Australia $t(1, 131.3) = 4.64$, $p = 0.015$, $M_{diff} = 3.44$. No other post-hoc comparisons approached significance, (all $t < 1.4$, all $p > 0.486$). Given differences across recruitment sites, institution was included as a control variable in all analyses.

2.4. Does attention bias variability for contamination-related stimuli account for variance in contamination fear above average attention bias?

While examination of the relationships between contamination fear with attention bias variability above average attention bias was predicated on a potentially significant association between average attention bias and contamination fear, it is possible that the small (though non-significant) amount of variance shared between average attention bias index measures and contamination fear ($r = 0.055$ and 0.051) could render the small associations between either TLBS measure non-significant. As such we proceeded with planned examination of whether attention bias variability for contamination-related stimuli is associated with significant variance in contamination fear beyond average attention bias index measures using a hierarchical regression. The control variables of institution (recruitment location) and response-time variability were entered at Step 1, $\mathrm{ABI_{Contam}}$ was entered at Step 2 and $\mathrm{TLBS_{Contam}}$ was entered at Step 3. As depicted in Table 3, at Step 1

[1] At the suggestion of an anonymous reviewer of this manuscript, we also computed an alternative measure of response time variability based on the same principles as the TLBS attention bias variability measure. This involved pairing of sequential trials and arbitrarily subtracting one from the other. The difference between these consecutive indices was then computed, the resulting value normalised, and the mean of the normalised values used as the alternative RTV measure. The correlation between the original computation method and this alternative version was $r(315) = 0.974$. No changes to the significance or the direction of any effects from the original analysis were observed when using this alternative RTV measure.

Table 2
Pearson's correlations between attention bias measures, response time variability and contamination fear.

	ABI_{Contam}	ABI_{Mit}	ABV-Mov_{Contam}	ABV-Mov_{Mit}	$TLBS_{Contam}$	$TLBS_{Mit}$	RTV
ABI_{Mit}	−0.067						
ABV-Mov_{Contam}	0.022	0.042					
ABV-Mov_{Mit}	−0.036	−0.103	0.377***				
$TLBS_{Contam}$	0.021	−0.102	0.423***	0.453***			
$TLBS_{Mit}$	0.001	0.031	0.325***	0.497***	0.703***		
RTV	−0.028	−0.038	0.149**	0.232***	0.257***	0.260***	
Padua-Contam	0.055	0.051	0.068	0.025	0.133*	0.147**	0.010

Note: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. See Table 1 key for variable names.

Table 3
Hierarchical regression with contamination fear (Padua inventory) as criterion variable and Institution (Step 1 - effects described above and in Table 1), Response Time Variability (RTV; Step 1), average attention bias index measure for contamination-related stimuli (ABI_{Contam}; Step 2), and trial-level bias score measure of attention bias variability for contamination-related stimuli ($TLBS_{Contam}$; Step 3) as predictor variables. Effects of institution described above and in Table 3.

	Predictor	Estimate	Stand. Estimate	SE	*t*	R^2	ΔR^2	*p*
Step 1	RTV	0.001	0.013	0.003	0.225			0.822
	Institution					0.038		0.017
Step 2	RTV	0.001	0.014	0.003	0.246			0.806
	ABI_{Contam}	0.007	0.039	0.011	0.700		0.002	0.485
						0.039		0.029
Step 3	RTV	−0.002	−0.028	0.004	−0.482			0.630
	ABI_{Contam}	0.006	0.034	0.010	0.605			0.546
	$TLBS_{Contam}$	0.023	0.157	0.009	2.733		0.022	0.007
						0.062		0.003

Note: Institution included as a covariate at each step, effects described in text and in Table 1.

Institution and RTV predicted significant variance in contamination fear. While Institution was observed to be a significant predictor $F(3,310) = 4.05$, $p = 0.008$, RTV was not $F(1,310) = 0.05$, $p = 0.822$. At Step 2, the addition of ABI_{Contam} did not explain significant additional variance. At Step 3 the addition of $TLBS_{Contam}$ did account for additional significant variance, being a significant unique predictor of contamination fear. This is consistent with the hypothesis that attention bias variability in relation to contamination related information accounts for additional variance beyond average biased attention toward contamination related information.

Does attention bias variability for mitigation-related stimuli account for additional variance in contamination fear above average attention bias?

To examine whether attention bias variability for mitigation-related stimuli is associated with additional shared variance with contamination fear (Padua subscale) beyond average attention bias index measures, the identical hierarchical regression to that described above was conducted with mitigation-related stimuli instead. The effect of Institution and RTV at Step 1 was identical to the previous model. At Step 2, the addition of ABI_{Mit} did not explain significant additional variance, with ABI_{Mit} not accounting for significant unique variance in contamination fear. At Step 3 the addition of $TLBS_{Mit}$ did account for additional significant variance, being a significant unique predictor of contamination fear. Effects are summarised in Table 4. The observed pattern of effects is consistent with the hypothesis that attention bias variability in relation to mitigation related information would account for additional variance beyond average biased attention toward mitigation related information.

To examine whether either or both $TLBS_{Contam}$ or $TLBS_{Mit}$ account for unique variance in contamination fear above the other, an exploratory follow-up analysis was conducted with two separate hierarchical regressions. Each included RTV and one TLBS measure in Step 1 with the other TLBS measure in Step 2. Examination of R^2 change values indicated that $TLBS_{Contam}$ did not account for additional variance beyond $TLBS_{Mit}$ $\Delta R^2(1311) = 0.002$, $p = 0.416$, and $TLBS_{Mit}$ did not account for unique variance beyond $TLBS_{Contam}$ $\Delta R^2(1311) = 0.006$, $p = 0.163$. This suggests that the majority of variance between these measures of attention bias variability and contamination fear was shared rather than independent.

3. Discussion

The present study aimed to determine if contamination fear was

Table 4
Hierarchical regression with contamination fear (Padua inventory) as criterion variable and Institution (Step 1 - effects described above and in Table 1), Response Time Variability (RTV; Step 1), average attention bias index measure for mitigation-related stimuli (ABI_{Mit}; Step 2), and trial-level bias score measure of attention bias variability for contamination-related stimuli ($TLBS_{Contam}$; Step 3) as predictor variables.

	Predictor	Estimate	Stand. Estimate	SE	*t*	R^2	ΔR^2	*p*
Step 1	RTV	0.001	0.013	0.003	0.225			0.822
	Institution					0.038		0.017
Step 2	RTV	0.001	0.014	0.003	0.248			0.804
	ABI_{Mit}	0.006	0.037	0.010	0.652		0.001	0.515
						0.039		0.030
Step 3	RTV	−0.002	−0.025	0.004	−0.436			0.663
	ABI_{Mit}	0.005	0.031	0.010	0.557			0.578
	$TLBS_{Mit}$	0.021	0.147	0.008	2.561		0.020	0.011
						0.059		0.004

Note: Institution included as a covariate at each step, effects described in text and in Table 1.

associated with attention bias variability for contamination-related and mitigation-related information in a large sample during the early stages of the COVID-19 pandemic. Interestingly, we found no support for an association between contamination fear and the standard average measure of attention bias for either contamination-related or mitigation-related stimuli. However, ABV for both contamination-related information and mitigation-related information showed a small but statistically significant correlation with contamination fear. These measures of ABV also accounted for significant unique variance above the standard average measures of attention bias, but given the negligible associations observed with the average measures, this in itself was not surprising.

Interestingly, neither the contamination nor mitigation measure of ABV accounted for significant unique variance in contamination fear above the other. This, and the observation that these ABV measures correlate at 0.70 suggests that the association between attention bias variability for contamination-related information or mitigation-related information and contamination fear largely constitutes shared variance. In the context of the current study this could suggest that the common factor between variability to both contamination and mitigation information with contamination fear is relatedness to virus concerns. More broadly, it is possible therefore that the magnitude of variability in attention bias characterised by variation between attentional vigilance and avoidance across trials is similar for contamination-related and mitigation-related of stimuli. This would tend to support the possibility that attentional processing of danger-relevant and mitigation-relevant stimuli in contamination fear are inextricably linked (Vogt et al., 2011). As such, it may not be necessary to assess both stimulus types to provide an indication of attention bias variability in contamination fear.

The present finding supports the hypothesis that attention bias variability is a feature of contamination fear. Importantly, this relationship was still present when controlling for response time variability. A number of researchers have highlighted the possibility that attention bias variability may be partially or fully the product of response time variability (Carlson et al., 2022; Kruijt et al., 2016), underscoring the importance of controlling for response-time variability when examining relationships involving ABV. Indeed a study by Carlson et al. (2022) in a high trait anxious sample (who also exhibited attention bias for threat) found that a significant relationship was indeed observed between ABV and anxiety symptoms, though was no longer significant when general response-time variability was controlled for. While the attenuated sample examined in the study (high trait anxious, with high average attention bias for threat) may have restricted the ability to detect associations, this finding nevertheless highlights the importance of controlling for response time variance when examining relationships involving ABV. In the current study response time variability did show a small but significant relationship with both the contamination and mitigation TLBS measures of ABV ($r = 0.257$, and 0.260 respectively), but did not correlate significantly with contamination fear ($r = 0.01$). Furthermore, the associations observed between ABV measures and contamination fear remained significant when controlling for response time variability. As such the present results are consistent with other studies where relationships between anxiety/emotionality and ABV remain significant after controlling for response time variability (Clarke et al., 2020).

While the current study is not alone in observing no association between the general tendency to align attention with contamination information (i.e. average attention bias) and contamination fear (Armstrong et al., 2012), it is worth considering why these theoretically related constructs showed no significant association. As with other studies (Chapman et al., 2019; Schmukle, 2005), the average attention bias index measure was observed to have low internal consistency ($r =$ 0.236 - 0.246), which would have diminished the ability to detect a relationship. Another possibility concerns the context in which participants completed the study. While the timing of the study meant that the broader environmental context may have been characterised by heightened objective risk of COVID-19 contamination, along with many visible reminders (masks, sanitization stations etc.), the immediate context in which participants were assessed (i.e. online – in their homes) likely diminished the salience of such threat. As highlighted by the findings of Vogt et al. (2011), it may be necessary to induce acute concerns regarding contamination in order to elicit a more general pattern of biased attention for contamination and mitigation-related information.

The current study was among the first to include two of the common approaches to assessing attention bias variability in a single study. Interestingly, the trial-level bias score measures were the only scales that showed an association with contamination fear, and incidentally showed the highest split-half reliability. By contrast, one of the few other studies to also examine both measures of attention bias variability found that the moving average measure was more strongly associated with anxiety and stress compared to the trial-level bias score measure (Clarke et al., 2020). As such, it will be important for future research to continue to incorporate both measures to permit evaluation of their relative sensitivity to emotion-linked attentional processes.

It is also relevant to note that while the results of the current study are consistent with a small association between ABV for contamination-related stimuli in contamination fear, they do not preclude the possibility that these results reflect a broader association between ABV for general threat stimuli and heightened anxiety/fear. To address this it would be necessary to also include a task assessing ABV for general threat stimuli and dispositional fear/anxiety, and then to assess whether ABV for contamination-related stimuli predicts unique variance in contamination fear above ABV for general threat stimuli, and conversely, that ABV for general threat stimuli accounts for unique variance in general fear/anxiety beyond ABV for contamination-related stimuli. Assessing the potential dissociability and specificity of relationships between ABV and aspects of emotional vulnerability will therefore be an important avenue for future research.

While the present study included a number of notable strengths, including a robust sample size and measures of attention bias for both contamination and mitigation-related information, some potential limitations should also be noted. The online delivery format allowed larger scale international data gathering and also represented a necessity of the conditions at the time of data collection, given restrictions on face-to-face research. Nevertheless, it is possible that completion of the study at home meant that participants were exposed to a greater number of distractors than in a lab environment. It will be important therefore to replicate the current findings in a lab-based setting that will permit greater control, and also the potential to incorporate manipulations, such as exposure to material that could elicit acute concern regarding contamination. Also, we opted for word stimuli due to its capacity to represent stimulus classes that may either differ in terms of personal representations across individuals (e.g. 'dirty') or are difficult to represent in pictorial stimuli (e.g. 'contagious'). Nevertheless, such symbolic stimuli may be less evocative than concrete pictorial representations of feared (or protective) stimuli. As such, it will be interesting for future research to examine whether the present results replicate with different stimuli. Given that the primary sample represented young adults, it is also relevant to consider whether the present results would generalise to a more diverse sample. As this population was known to be at lower risk of severe health complications resulting from COVID-19 infection, this could have contributed to attenuated concerns regarding infection.

It is also relevant to consider whether the ordering of task delivery may have impacted observed effects at all, and in particular whether the measure of RTV in a separate block may have accurately captured patterns of RTV that also operate in the probe task. As highlighted by the findings of Carlson and Fang (2020), measures of attention bias variability can change across blocks of the dot probe task and as such, it is possible that RTV may also change over time. While the present data cannot directly address the potential ordering effect, examination of the internal consistency of the RTV measure itself can inform whether early

RTV is reflective of later RTV on this task. The high internal consistency in the first/second half split ($r = 0.905$, Spearman-Brown corrected = 0.947) provides some reassurance that RTV was generally stable, at least within this task. Thus, while it will be important for research to confirm the relative stability and potential practice/fatigue effects associated with the measurement of RTV in future studies, the high internal consistency observed could suggest that RTV may be relatively stable.

The absence of associations involving the standard average attention bias measure in the current study could be considered unsurprising given the very low internal consistency observed. The split half associations of the average attention bias index measures, such as ABI_{Contam} ($r = 0.236$) and ABI_{Mit} ($r = 0.246$), as well as the moving average attention bias variability measures like ABV-Mov_{Contam} ($r = 0.254$) and ABV-Mov_{Mit} ($r = 0.177$), was observed to be very low, which aligns with findings from previous studies (e.g. Chapman et al., 2019; Schmukle, 2005). This low internal consistency raises important considerations about the reliability and interpretability of these measures within the context of our study and indeed others. Such elevated measurement error can impact research findings in a range of ways. One of these is the increased likelihood of spurious effects registering as significant within small samples (Loken & Gelman, 2017; Schmidt et al., 2014). Specifically, as scores obtained from measures with low reliability produce larger distributions, this can increase the likelihood of sampling error withing low-powered studies which in turn produce false positive results. Given that much of the research examining patterns of biased attention in contamination fear has been conducted with reasonably small samples, effects reported from such low powered studies should be regarded with appropriate scepticism.

In addition to producing potentially spurious effects in small samples, low reliability can also obscure effects in larger samples due to elevated noise (Armstrong, 1998). In the context of the present results, the observed low reliability of the average attention bias measures suggests that it is not possible to draw definitive conclusions regarding the absence of effects involving average attention bias and contamination fear. To more definitively address questions regarding the presence of average attention bias for threat in contamination fear (and indeed other dispositional traits) it will be critical to replicate studies using attention bias assessment tasks with higher internal consistency. This may include the use of eye tracking measures of selective attention which have been shown to produce considerably higher internal consistency for emotional stimuli ($r = 0.72$ - 0.84; Sears et al., 2019). Other recent research has sought to develop probe task variants using both static stimuli and more dynamic video stimuli that show higher reliability. Across four studies, Grafton et al. (2021) showed encouraging results in relation to both internal consistency ($r = 0.67$ - 0.97) and test-retest reliability ($r = 0.67$ - 0.71). The further refinement and implementation of such approaches will therefore be important to advancing knowledge regarding attention bias in contamination fear, and individual difference dimensions more broadly.

The split-half reliability of the TLBS measures employed in the current study fell within the marginal range for the uncorrected values ($r = 0.582$ and 0.609) and within the acceptable range when using the Spearman-Brown corrected values ($r = 0.736$ and 0.757). While these values are substantially higher than the average attention bias measures they would nevertheless be considered relatively low in the context of individual differences research. While the larger sample provides reassurance that the present results are unlikely to be the product of sampling error, a noted effect of lower reliability is biasing correlation coefficients downward (Fan, 2003; Padilla & Veprinsky, 2012). It is possible therefor that the true magnitude of the relationships observed may well be larger. While such tasks may therefore be suitable for addressing theoretical questions regarding the role of cognitive processes in emotion when using large samples as in the current study, improved reliability will be critical for examining such relationships in clinical contexts.

In considering the results of the present study, it is also important to acknowledge the small effect sizes observed in the relationships between attention bias variability measures and contamination fear, with correlations of $r = 0.133$ and 0.147. While statistically significant, the practical significance of such effects, particularly in clinical contexts may well be limited, even if the true values of these relationships have been attenuated due to measurement error. This study was primarily designed to investigate population-level patterns of attention bias variability in a context where contamination fear was generally heightened. The findings, therefore, contribute to a foundational understanding of these patterns rather than directly informing clinical interventions or decision-making. The small magnitude of these effects suggests that while statistically significant, the potential contribution of attention bias variability to contamination fear may be very small. As such, these results should be interpreted with caution, with further research required examining clinical samples in particular required to determine the potential applicability to clinical contexts.

In summary, the present study examined whether attention bias variability for contamination-related and mitigation-related stimuli was associated with contamination fear during the early stages of the COVID-19 pandemic. While average attention bias towards either stimulus type was unrelated to contamination fear, trial-level bias score measures of attention bias variability for both contamination-related and mitigation-related stimuli showed a small positive relationship with contamination fear. This finding is consistent with the hypothesis that attention bias variability is a feature of contamination fear and invites further research into the presence of and conditions under which bias variability may operate in contamination fear.

CRediT authorship contribution statement

Patrick J.F. Clarke: Data curation, Formal analysis, Methodology, Project administration, Software, Supervision, Writing – original draft, Writing – review & editing. **Elise Szeremeta:** Formal analysis, Investigation, Project administration. **Bram Van Bockstaele:** Conceptualization, Investigation, Project administration, Writing – review & editing. **Lies Notebaert:** Data curation, Investigation, Methodology, Project administration, Software. **Frances Meeten:** Data curation, Investigation, Methodology, Writing – review & editing. **Jemma Todd:** Data curation, Investigation, Methodology, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

A link to the data on Open Science Framework are available in the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.brat.2024.104497.

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